Food and Agriculture Organization of the United Nations

Strengthening Emergency Preparedness and Response to Food Crisis in South Sudan (SEPAREF)

UTF/SSD/027/SSD

INTEGRATED PEST MANAGEMENT PLAN

This document is intended to use solely for the purpose of FAO projects disclosure

July, 2024

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1. Introduction

The African Development Bank is preparing to support the Government of South Sudan through FAO as a lead implementing agency in the implementation of Strengthening Emergency Preparedness and Response to Food Crisis Project in South Sudan with the objective of increasing access to quality foundation seed to enhance crop production and productivity through organizes famers seed producers and Seed companies and introduction of improve varieties and farming technologies. The project will support technology generation and dissemination by supporting the strengthening and scaling up of Farmers seed producers and seed companies.
In south Sudan the project will focus on supporting crop seed production (M Sorghum, Rice , Cowpea) and fodder.

- Where projects involve recourse to pest management measures, FAO will give preference to integrated pest management (IPM) or integrated vector management (IVM) approaches using combined or multiple tactics.
- The project will procure o pesticides / Fungicides product only for seed treatment, the IMP will be promoted as the best option for pest management.

ESS3 applies to all Bank lending. Even if Bank lending for pesticides is not involved, an agricultural development project may lead to substantially increased pesticide use and subsequent environmental problems.

The ESS3 also states that:

- The following additional criteria apply to the selection and use of such pesticides: (a) they will have negligible adverse human health effects; (b) they will be shown to be effective against the target species; and (c) they will have minimal effect on non-target species and the natural environment. The methods, timing, and frequency of pesticide application are aimed to minimize damage to natural enemies. Pesticides used in public health programs will be demonstrated to be safe for inhabitants and domestic animals in the treated areas, as well as for personnel applying them; (d) their use will take into account the need to prevent the development of resistance in pests; and (e) where registration is required, all pesticides will be registered or otherwise authorized for use on the crops and livestock, or for the use patterns for which they are intended under the project.

Procedures for assessing pest management issues are described in the Bank's policy on environmental assessment (ESS 4 *Environmental Assessment*), of the supporting Bank Procedure (EES 4.01 - *Application of EA to Projects Involving Pest Management*). When significant pest management issues are associated with a project, ESMP developed during the EA process should include a pest management plan.

A pest management plan is a comprehensive plan, developed when there are significant pest management issues such as (a) new land-use development or changed cultivation practices in an area, (b) significant expansion into new areas, (c) diversification into new crops in agriculture, (d) intensification of existing low-technology systems, (e) proposed procurement of relatively hazardous pest control products or methods, or (f) specific environmental or health concerns (e.g., proximity of protected areas or important aquatic resources; worker safety).

A pest management plan reflects the policies set out in ESS3. The plan is designed to minimize potential adverse impacts on human health and the environment and to advance ecologically based IPM.

2. Requirements

2.1 General

FAO promotes Integrated Pest Management (IPM) as a pillar of sustainable agriculture-IPM means the careful consideration of all available pest control techniques and subsequent integration of appropriate measures that discourage the development of pest populations and keep pesticides and other interventions to levels that are economically justified and reduce or minimize risks to human and animal health and/or the environment. IPM emphasizes the growth of a healthy crop with the least possible disruption to agro-ecosystems and encourages natural pest control mechanisms.

2.2 Pest Management Framework

If provision or use of large volumes of pesticides is foreseen, a Pest Management framework (PMF) needs to be prepared to demonstrate how IPM will be promoted to reduce reliance on pesticides, and what measures are taken to minimize risks of pesticide use. Such a PMF needs to be an integral part of the Environment and Social Commitment Plan. A Framework has been developed instead of a Plan as project locations are still not confirmed. This Framework will be used to guide the development of Pest Management Plans that address area-specific concerns once project locations are known.

2.3 Selection of pesticides

If after having considered available IPM approaches, pesticide use is deemed to be justified, then careful and informed consideration should be given to the selection of pesticide products. Factors to be taken into account include hazards and risks to users, selectiveness and risk to non-target species, persistence in the environment, efficacy and likelihood of development or presence of resistance by the target organism. Minimum environment and social analysis are needed.

FAO does not maintain a list of permitted or non-permitted pesticides because many locally specific conditions govern which pesticides may be used. However, in line with the provisions of the FAO/WHO International Code of Conduct on Pesticide Management and relevant multilateral environmental agreements that include pesticides, the following list of criteria will need to be met in order for a pesticide to be considered for use in an FAO project:

a. The product should be registered in the country of use, or specifically permitted by the relevant national authority if no registration exists. Use of any pesticide should comply with all the registration requirements including the crop and pest combination for which it is intended.

b. Users should be able to manage the product within margins of acceptable risk. FAO will not supply pesticides that meet the criteria that define Highly Hazardous Pesticides (HHPs) **3**. Pesticides that fall in WHO Hazard Class 2 or GHS Acute Toxicity Category 3 can only be provided if less hazardous alternatives are not available and it can be demonstrated that users adhere to the necessary precautionary measures4.

c. Preference should be given to products that are less hazardous, more selective and less persistent, and to application methods that are less hazardous, better targeted and requiring less pesticides.

d. Any international procurement of pesticides must abide with the provisions of the Rotterdam Convention on the Prior Informed Consent (PIC) Procedure.

2.4 Steps to planning PMF

The following six steps will assist in effective pest management planning:

1. Understand the pest issues
2. Develop a draft pest management plan
3. Consultations
4. Finalize and implement the plan
5. Monitoring
6. Evaluate and review the overall results

2.5 Principles of Pest Management

Eight principles of pest management are suggested to follow as common basis for the management of pest throughout project area. The consideration of all these principles is critical to the success of any pest management activity, regardless of scope and scale. These are:

1. Integration: Pest management in Agriculture production is an integral part of managing natural resources and agricultural systems.
2. Public awareness: Public awareness and knowledge of pest must be raised to increase the capacity and willingness of individuals to participate in control.
3. Commitment: Effective pest management requires shared responsibility, capability, capacity and a long-term commitment by farmers/ operators/ processors/ landowners/ managers, the community, industry groups and government. Those that create the risks associated with pest introduction or spread and those that benefit from the pest management should help to minimize the impacts of pest and contribute to the costs of management.
4. Consultation and partnership: Consultation and partnership arrangements among the users, local communities, industry groups, government agencies and local governments must be established to achieve a collaborative and coordinated approach to management.
5. Planning: Planning for pest management should be based on risk management to ensure that resources target the priorities identified at local, regional, and national levels.
6. Prevention and early intervention: Preventive pest management is generally more cost-effective than other strategies and is achieved by preventing the spread of pest species, and viable parts of these pests, especially by human activity early detection and intervention.
7. Best practice: Pest management must be based on ecologically and socially responsible practices that protect the environment and the productive capacity of natural resources while minimizing impacts on the community. It should balance feasibility, cost-effectiveness, sustainability, humaneness, community perceptions, emergency needs and public safety.
8. Improvement (research, monitoring and evaluation): Research about pest and regular monitoring and evaluation of control activities.
9. Needed to make evidence-based decisions and improve pest management practices.

Transition to a PMP program requires a diverse, action-oriented PMP Committee. This PMP Committee will be an environmentally conscious committee lead by the Project manager at PMU, A representative of the county agriculture Office and Farming Group will be the members of this committee. The leader of this team should be familiar with pests, pesticides and pesticide regulations. This arrangement is appropriate, because implementation of an IPM program can be tracked as a performance indicator.

PMP leadership is guided by pest management principles and environmental issues. Leadership with such academic background and experience qualifies to serve as an authority to supervise PMP implementation. Other team members will include Environmental, Agriculture Extension, agronomists, crop protection experts (entomologists, pathologists), aquaculture experts, health officer and livestock officer.

The use of pesticides must be guided by the principles of cost efficiency, safety to humans, the bio-physical environment and effectiveness in controlling the pests. Pesticides selection will be made in accordance with the African Development Bank guidelines and FAO for the selection of pesticides that requires:

i. Pesticides requiring special precautions should not be used if the requirements are not likely to be met;
ii. Pesticides to be selected from approved list, taking into consideration of: toxicity, persistence, user experience, local regulatory capabilities, type of formulation, proposed use, and available alternatives;
iii. Type and degree of hazard and availability of alternatives; and the following criteria will be used to restrict or disallow types of pesticides under Bank loans:
    - **a. Toxicity**: acute mammalian toxicity, chronic health effects, environmental persistence and toxicity to non-target organisms;
    - **b. Registration status** in the country and capability to evaluate long-term health and environmental impacts of pesticides.

The Integrated Pest Management and Monitoring Plan (IPMP) is to be developed from the impacts and mitigation measures identified at the implementation stage based on the principles mentioned in this chapter and also the available techniques for Agriculture farms are described in Chapter 3. The IPMP should include impacts from application of chemical as well as non-chemical pesticides. The reason why chemical pesticides are included is that in the initial stages of implementation of the IPM, chemical pesticides , will still be used but will be gradually phased out as the IPM gets established.

Training programs on various aspects of the pest and disease management and judicious use of chemical pesticides have to be organized by the pest management team

Training modules for pest management in farms and nurseries should be developed. Following training programs will be provided, farmers' training, Pesticides dealers' training, Agriculture extension personnel including staff training, and Local service providers training.

To initiate the promotion of IPM and sound pesticide use will be effective by organizing awareness program involving Farmer's Groups and different stakeholders. Awareness will be raised through demonstrations, discussion meetings, dissemination of information about pest arrival, distribution of leaflet, booklet, etc.

3. Integrated Pest Management Plan (IPMP)

The IPMP should include impacts from application of chemical as well as non-chemical pesticides. The reason why pesticides are included is that in the initial stages of implementation of the IPM, pesticides will still be used the last resort, and will be gradually phased out as the IPM is established.

When coming up with the IPMP, the following steps should be considered and documented:

- Identify the main pests affecting crops in the areas of the project / Country, assess the risks to the operation, and determine whether a strategy and capacity are in place to control them.

- Where possible, apply early-warning system for pests and diseases (i.e., pest and disease forecasting techniques).
- Select resistant varieties and use the cultural , mechanical and biological control of pests, diseases and weeds to minimize dependence on pesticide (chemical) control options. An effective IPM regime should:
    - Identify and assess pests, threshold levels, and control options (including those listed below), as well as risks associated with these control options.
    - Rotate crops to reduce the presence of insects, disease, or weeds in the soil or crop ecosystems.
    - Support beneficial bio-control organisms—such as insects, birds, mites, and microbial agents—to perform biological control of pests (e.g., by providing a favorable habitat, such as bushes for nesting sites and other original vegetation that can house pest predators and parasites).
    - Favor manual, mechanical weed control and/or selective weeding.
    - Consider using mechanical controls—such as traps, barriers, light, and sound to kill, relocate, or repel pests.
    - Use pesticides to complement these approaches, not replace them.
    - Prior to procuring any pesticide, assess the nature and degree of associated risks and effectiveness, taking into account the proposed use and the intended users.

3.1 The Objective of IPMP

The purpose of the IPMP is to ensure that the identified impacts related to application of pesticides are mitigated, controlled or eliminated through planned activities to be implemented throughout the project life. The IPMP also provides opportunities for the enhancement of positive impacts. The IPMP gives details of the mitigation measures to be implemented for the impacts; and the responsible institutions to implement them.

Implementation of the IPMP may be slightly modified to suit changes or emergencies that may occur on site at the time of project implementation. The plan therefore should be considered as the main framework that must be followed to ensure that the key potential negative impacts are kept minimal or under control.

In this regard, flexibility should be allowed to optimize the implementation of the IPMP for the best results in pest management. The IPMP consists of generic or typical environmental impacts that are derived from the site investigations, public consultations and professional judgment. This is because the specific and detailed impacts cannot be predicted without details for the project design and construction activities as well as the specific project locations. The IPMP will however, provide guidance in the development of more detailed IPMP's, once the project design and construction details are known. Site specific Integrated Pest Management and monitoring plans will depend on the scope of identified major impacts to be addressed in the implementation of the Project.

3.2 Pest Monitoring Plan

Successful implementation of the Integrated Pest Management Plan in the project locations will require regular monitoring and evaluation of activities under taken by the farmers to be involved in the project. The focus of monitoring and evaluation will be to assess the build-up of IPM capacity among the farmers and the extent to which IPM techniques are being adopted in Agriculture production, and the economic benefits that farmers derive by adopting IPM. It is also crucial to evaluate the prevailing trends in the benefits of reducing pesticide distribution, application and misuse.

Indicators that require regular monitoring and evaluation during the programme implementation include the following:

1. Number of farmers engaged in IPM capacity building in the project locations
2. Number of farmers who have successfully received IPM training in IPM methods
3. Number of trainees practicing IPM according to the training instructions
4. Number of women as a percentage of total participating in IPM and successfully trained
5. Number of youth as a percentage of total participating in IPM and successfully trained
6. Number of farmers as a percentage of total applying IPM

4. Steps in Setting up IPM

4.1 Identify the implementation team

Transition to a PMP program requires a diverse, action-oriented PMP Committee. This PMP Committee will be an environmentally conscious committee lead by the Project Director at PMU, DPP, Representative of the County Agriculture Department(CAD) and Farming Group will be members of this Committee. The leader of this team should be familiar with pests, pesticides and pesticide regulations. This arrangement is appropriate, because implementation of an IPM program can be tracked as a performance indicator.

PMP leadership is guided by pest management principles and environmental issues. Leadership with such academic background and experience qualifies to serve as an authority to supervise PMP implementation. Other team members include Environmental, Agriculture Extension , agronomists, crop protection experts(entomologists, pathologists), aquaculture expert, health officer and Livestock officer.

4.2 Decide on the scale of implementation

To determine the scale of implementation, a strategic approach will be taken. IPM will be clearly defined and discussed by the PD as is done for all other development projects. A representative of the CAD offices must attend these meetings to help explain the IPM approach and give examples of similar documented success studies. Through these discussions comprehension will be achieved, and potential objections will be addressed with successful practical examples.

4.3 Review and set measures objectives for the PMP

The PMP Committee will set measurable objectives and refine the IPM indicators relevant to their Counties / State; and determining factors such as:

- When the IPM program will start
- How much it will cost
- What will be accomplished by choosing IPM
- How success shall be monitored

The determination of above must be done prior to IPM implementation. Additionally, measurable goals will be set, to track:

- Pest management costs;

- Monitoring of pest activity before and after implementation of the IPM program;
- Number of calls related to pest problems and toxic chemical use reduction.

Furthermore, the time when the shift to IPM will occur must be discussed and agreed upon prior to implementation. The initial step will be to establish an implementation timeline that includes time to execute all of the steps outlined in the implementation plan. It is imperative to include time to organize the administration of the IPM and conduct any farmer training as well as manage the IPM process.

The IPM Committee will gather information on previously implemented or currently being implemented IPM programs; the time it took to develop them and how successful they have been. They will obtain the budgetary and any technical information for the previously implemented IPM programs and analyze the elements to establish lessons to learn. Field visits to currently running programmes will be conducted to get a practical insight.

Reduced pesticide use is the substantive yardstick in measuring an IPM's ability to create a safer environment. Baseline study will be conducted and therefore an information database that includes annual quantities of pesticides used will be designed to enable comparative analysis to the previous years. The goal will be a downward trend over time or ideally, a specific reduction amount, ultimately leading to a scant usage of highly toxic pest control chemicals.

4.4 Analysis current housekeeping, maintenance and pest control practices

While preparing to make a transition to IPM, the PMP Committee will familiarize itself with the organization's current policies and practices with respect to structural maintenance, sanitation and pest control. Occasionally, current practice may be consistent with IPM principles. Familiarization will provide the flexibility necessary to adapt to, and prepare for the necessary changes.

Structural maintenance is arguably the most efficient way to keep pests out of a facility because it physically stops pests from entering wherever possible. Structural maintenance will therefore be a regular part of the IPM. Cracks, crevices or other unnecessary openings in the building exterior that can be used by pests as harborage areas or entry points regardless of size, will be sealed appropriately. Sanitation deprives pests of food and water. A sanitation plan must therefore be accounted for in the development of an IPM. Staff must be provided with special sanitation training.

4.5 Establish a system of regular IMP inspection

PMP's central focus is regular facility inspections. Such inspections are the "life blood" for a continuous cycle of IPM activities that may or may not include chemical treatments. Activities will include:

a) Routine Inspections
b) Pest Identification
c) Selection of Control Methods
d) Monitoring and Evaluation

IPM inspections must emphasize on the four "zones" of pest activity:

a) Entry points
b) Water sources
c) Food sources
d) Harborage areas.

During inspections, all existing pest issues and potential problem areas, inside and outside, must be noted for follow-up.

For in-house IPM programs, the greatest inspection challenge will be establishing routine, proactive surveillance by trained specialists. To ensure this is done, the EMC or an independent consultant will conduct inspections and audits twice a year.

4.6 Define the treatment policy selection

A clear written policy on how the facility will respond to pests, when they appear,must be developed. Included in the policy will be definitions of both non-chemical and chemical treatment options and the sequence or prioritization in which they will be considered. It should be unequivocal on when and where chemical treatments are appropriate. Finally, it should include an "approved materials" list to ensure informed choices when chemical treatments are applied.

The key to an effective IPM is to correctly identify pests that have invaded the area before. Due to pest behavior variations from one species to the other, the appropriate response will vary accordingly.

Once the pest is identified and the source of activity is pin-pointed, the treatment policy will call for habitat modifications such as exclusion, repair or better sanitation. These counter measures can drastically minimize pest presence before chemical responses are considered. Additional treatment options—chemical and non-chemical can then be tailored to the biology and behavior of the target pest.

The final step in the pest response cycle is Monitoring. The information gained through on-going monitoring of the problem will facilitate determination of supplemental treatment options if required.

4.7 Establish communication protocols

Communication protocols must be developed to assist environmental services, facility maintenance, facility management and service providers. IPM is a cooperative effort and therefore effective communication between various parties is essential for success. PMP Committee and fish farmers must document pest sightings.

The PMP Committee will make recommendations and notify CAD for pesticide treatments. They will also communicate with the maintenance team to make the necessary repairs.

4.8 Develop Agriculture farmer training plans and policies

The Agriculture Farmer Groups will serve as a pool of "inspectors" charged with reporting pest sightings to expedite response times and help limit the scope of new infestations. Training sessions will be conducted to acquaint farmers with IPM principles and their responsibilities for the success of the IPM program.

4.9 Track progress and reward success

Measurable objectives set at the beginning, must be measured against the IPM program's performance at least once a year. Documentation to facilitate the evaluation process is as follows:

- ✓ Detailed description of the parameters and service protocols of the IPM program, stating the ground rules;
- ✓ Specific locations where pest management work was performed;
- ✓ Dates of service;
- ✓ Activity descriptions, e.g., baiting, crack-and crevice treatment, trapping, structural repair; hygiene and
- ✓ Log of any pesticide applications, including:
- Target pest(s);

- The brand names and active ingredients of any pesticides applied;
- PCB registration numbers of pesticides applied;
- Percentages of mix used in dilution;
- Volume of pesticides used expressed in kilograms of active ingredient;
- Applicator's name(s) and certification identity (copy of original certification and re-certification should be maintained);
- Facility floor plan on which all pest control devices mapped and numbered;
- Pest tracking logs (sightings and trap counts);
- Action plans, including structural and sanitation plans, to correct any pest problems;
- Pest sighting memos for IPM Committee to use in reporting pest presence to County Agriculture department Executive Committee (CAEC); and
- Using these records, and the goals of the IPM program (increased efficacy, lower costs and reduced pesticide use), the IPM Committee must see:
    - Fewer pest sightings and farmer complaints;
    - Lower monitoring-station counts over time;
    - Lower costs after the first 12-18 months, once IPM's efficacy advantage has had time to take effect; and
    - Downward trend in volume or frequency of chemical pesticide usage
    - Reduced pest infestations on the fishes

IPM is a team effort. Therefore, the PMP Committee will track and report the program's successes following each evaluation; and encourage good practices by recognizing farmers who played a role. Communicating the success of the program in reducing toxic chemical use and exposure, reducing pest complaints and lowering costs will help farmers to understand the purpose of the program and appreciate its success. The more they understand, the more likely they will participate willingly in helping expand and institutionalize the IPMP.

After the program has been in place for long enough to show significant results, it is recommended for the PMP Committee to work with PPD to publicize successes more broadly and to demonstrate the environmentally responsible approach to effective pest management and control. PMP Committee and Plant protection Department (PPD)will lead theexample by sharing success with other stakeholders.

Table 1: Integrated Pest Management and Monitoring Plan

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
	1. Positive Impact of Chemicals Pesticides						
1.	**Increase in Crop Yield**	Minimising Loss in quantities and qualitative of harvest due to pest attack and contaminations.	Implement a long term IPM programme to sustain productivity and combat negative effects of chemical pesticides	PPD-PMU farmers participation	IPMP	MAFS PMU	Annually
1.2	**Increase in economic growth**	Minimizing damages and crop losses (quantitative and qualitative)		PPD – PMU Farmers	IPMP	MAFS	Annually
	2. Negative impact of Pesticides						
2.	**Depletion of organic soil nutrients**	• Persistent use of chemical pesticides	• Apply soil conditioning measures which include IPM	Farmers	IPMP	PPD – PMU PIU	Quarterly
2.1	**Poisoning of non-target species including natural enemies and bio-pesticides**	• Lack of knowledge of chemical pesticides potency • Lack of knowledge and understanding of bio-pesticides. • Equipment malfunction • Use of wrong type of equipment • Wrong time and method of application (Spraying)	• Regulate registration and use of pesticides, application equipment and protective gears. • Train end users of pesticides on standard operation procedures for pesticides use	Regulatory authority Agro-input dealers Farmers and Farmer Organizations Public and private extension service providers PPD - PMU	IPMP	MAFS PMU PPD	Quarterly

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			and application as well as appropriate IPM practices. • Supervise and control use of chemical pesticides so that only approved and recommended ones are used • Provide PM equipment • Regularly maintain and clean equipment as recommended by supplier • Dispose old equipment as recommended by manufacturer. • Provide recommended protective gear • Use recommended and appropriate protective gear				

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			• Conduct trainings in IPM				
2.2	**Adulteration**	Lack of controls and enforcement of regulation	Strengthening regulatory capacities for inspection, sampling and testing	Pesticide regulatory authority, Manufactures, Suppliers and/or Distributors (agro-input dealers). Pesticides transporters, suppliers and research station	• Packaging, labelling and storage standards • Product specification • EMA 2008 • Pesticides regulation	MAFS PPD PMU	Quarterly
2.3	**Health and safety risks**	• Exposure to Pesticides • Chemical pesticides misuse (over/under use) due to lack of appropriate knowledge • Accidental or intentional poisoning due to Improper labelling or storage, frustration, social pressures	• Provide protective clothing and ensure it is used • Train farmers in proper pesticides handling. • Routine medical examination	• Agro-dealers • Transporters • Farmers • PMU • PIU	• Pesticides regulation • Labour regulation • PPD • SMAF • CAD	Ministry of Labour MoH MAFS SMAF CADPPD PMU	Annually
2.4	**Water, soil and environmental pollution**	• Lack of understanding of target location/field where pesticides is to be applied. • Inappropriate building for storage of pesticide • Cleaning of equipment	• Environmental Impact Assessment on storage and use. • Construction suitable warehouse	• Pesticides transporters and suppliers • Agro-input dealers • PMU • PIU • Farmers	Water pollution standard Pesticides equipment recommendation	PPD – CAD Department of Environmen t Ministry of Water	Quarterly

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
		• Disposal of remains of pesticides • Disposal of Containers and equipment • Lack of understanding of target location/field where pesticides is to be applied.	and bio-beds draining channels and draining dams • Use chemical remains to re-spray. • Clean equipment in one designated place • Use plants such as water lilies to absorb waste pesticides. • Take regular stock of pesticides • Use IPM Training of farmers, on storage, use and disposal of chemicals				
	Improper storage of pesticides by suppliers/distributers	Wrong shelving or Stacking	Routine inspection and inventory checks	Agro-dealers Farmer Organizations	PPD regulations Manufacturers guidelines	PPD ,CAD	Half Yearly
		• Inadequate storage Space	Provide adequate and separate storage space for Pesticides	Agro-dealers Farmer Organizations	• PPD regulation	PPD,CAD	Half yearly
		• Bad housekeeping					

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
		• Multipurpose use of warehouse					
	Improper use of pesticide application equipment	Multi-purpose use of equipment or pesticides	Control use of equipment and pesticides • Training on equipment use and management. • Integrated pest management(IPM)	Farmers and producer organizations	Pesticides Regulation	PPD -CAD	Quarterly
2.8	**Pesticides resistance in pest**	Lack of appropriate knowledge in pesticides application rates - under dose or overdose	Train farmer on safe use and application method of pesticides	PMU PIU Farmers	Pesticides regulation	PPD CAD	Yearly
	3. Positive impact of biological control						
3.	**Reduce environmental and health risk**		Promote environmental and health benefits of biological controls to the farmers to appreciate the advantage	PMU - PIU	EMA 2019(draft)	MOEF MAFS	Quarterly
3.1	**Reduction in time spent on application of chemical pesticides**		Inventory of indigenous biological control method and create awareness to the farmers to enhance knowledge	PMU PIU	IPMP	MAFS	Annually
3.2	**Adoption of improved varieties**	Sometime rural farmers may resist introduction of new	Create awareness on the new improved seed	PUM	IPMP	MAFS	Quarterly

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
	resistant or tolerant to pests and diseases	varieties and stick to traditional varieties	varieties , which are resistant to pest and will reduce the use of pesticides	PIU			
3.4	**Increase in soil fertility and reduction of soil erosion**	Tree cover as biological control of pest will result in increase in soil stability and reduction of erosion		PMU PIU	IPMP	MAFS	
	4. Negative Impact of Biological Control						
4.1	**Damage on other unintended crops**	Wrong application of method by farmers	• Apply biological control carefully with good knowledge • Train farmers on the appropriate application of biological control agent. • Training on integrated pest management	PMU – PIU Famers and farmer organizations	IPMP	MAFS	
4.2	**Risk of damage to crops**	The slowness of biological agents to act may frustrate IPM programmes as farmers are used to the rapid result of chemical pesticides	• Train farmers on the benefits of biological control as well as IPM	PMU - PIU	IPMP	MAFS	
	5. Positive Impacts Of Mechanical Methods						

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
5.1	**Reduction in time spent on fields managing and controlling pests**		Regularly services equipment and machinery maintain their efficiency				
	6. Negative Impact of Mechanical Methods						
6.	**Damage to Crops**	Use of heavy and spacious automated machinery					
4.5	**Health and safety Risk**	Personnel operating farm machinery may be exposed to accidents and sharp blades during farm operations and maintenance of the machinery.	• Promote IPM approach • Provide protective gears • Train farmers on machinery operation		• IPMP	MAFS Ministry of Labour	Annually
4.6	**Air pollution**	Generation of dust and release of carbon dioxide by farm machinery	• Servicing of farm machinery • Controlling of machinery speeds during operation to reduce generation of dust	PMU- Farmers	SSEMA (2019) Draft	DE MAFS	Quarterly
4.7	**Soil contamination**	Fuel and oil leaks from farm machinery and spills from discarded waste oil containers	• Discarding waste oil container in Approved disposal sites	PMU	SSEMA2019(Draft)	DE	Quarterly

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			• Vehicle Servicing and fuel /oil storage areas				
4.8	**Health and Safety risk**	Snake bites, hippo or crocodile attacks , sesi flies,scopoyong	• Provide protective clothing to workers • Train farmers on proper handling and operation of equipment • Promote IPM method	PMU MAFS Farmers	N/A	PPD - MAFS	Annually
4.9	**High costs for labour**	Employment of a lot of labour requires considerable amount of money to pay as wages	Conduct sensitisation and awareness campaigns in the project implementation area for farmers to adopt IPM as a sustainable method of managing a pest	PMU – Farmers	N/A	MAFS	Annually
4.10	**Increase in time spent managing pests**	Use of hoes and slasher requires long times to be spent by farmers to control pest in the fields	Community sensitisation and awareness campaigns for farmer to adopt IPM as a sustainable method of managing pest	Farmers	N/A	PMU	Annually
	7. Positive Impacts of IPM						
5.	**Increase in Agriculture yields**	Non chemical methods are generally slow	Train farmers in timely and appropriate use of	PPD - PMU	IPMP	MAFS	Annually

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			pest management techniques to protect (crops, vegetable and vegetables) from other pest and crop damage.				
5.1	**Contribution to food security**	Non chemical methods are generally slow	• Train pesticides marketers in selection and handling of approved pesticides • Train farmers in the appropriate application of the various IPM practices • Educate Farmers on Preservation techniques and timeframes of different integrated pest management options	PMU - MAFS	IPMP	MAFS	Annually
5.2	**Saving in foreign exchange**	Banned chemicals	• Train pesticides suppliers in selection of appropriate pesticides to be eligible for supplying to	PMU MAFS	Pesticides Regulation	PPD , CAD MAFS	Quarterly

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			project(MAFS – PMU) • Train farmers in the appropriate application of the various IPM practices				
5.3	**Contribution to offsetting rural/urban migration**	Banned chemicals	• Enforce regulation prohibiting importation of banned pesticides • Educe farmers on harmful consequences banned chemical pesticides • Assist local communities to establish cooperatives and market produce to potential markets for additional income	PMU Farmers	Pesticide regulation	MAFS PPD CAD	Quarterly
5.4	**Improved environmental protection**		• Enforce regulation prohibiting importation of	PMU	IPMP	PMU	Annually

S/N	Potential Issues/ Concerns	Cause of Concern	Control/Mitigation Measures	Responsible person/institution and cost per year per county	Standards/regulation	Monitoring institution	Monitoring frequency
			banned chemical pesticides • Educate farmers on harmful consequences of banned chemical Pesticides.				

Project Beneficiaries:
The project is expected to benefit all the famers in south Sudan. However this plan will be implemented in specific areas expected project location / counties, the primary beneficiaries will be affected farmers, pastoralists and households that targeted for this project

Pesticide Policy, Legal, International Requirements and Guidelines:
The following legal instruments provide guidance and regulation when implanting project the use pesticides in south Sudan. Also include are international conventions and guidelines that south Sudan is in process to be as a signatory to about pesticide use.

Stakeholder Engagement:
MAFS /FAO will present this IPMP as draft to identified stakeholders as part of public consultation and more specifically to seek input from the stakeholders on potential impacts and mitigation measures of the Project.

IPMP Disclosure:

This IPMP will be disclosed in accordance with ESS 10 disclosure standards on the Website of MAFS and News Paper and forwarded to the Banks for disclosure at its public information center (PIC) of the country at the Banks external website. This IPMP will also be disclosed in the project areas and made accessible to the beneficiaries.